SUZANNE CORY
High School

Name: _____ Teacher : _____

2023 MATHEMATICAL METHODS UNIT 2

Assessment Task: Probability

Writing time: 38 minutes

QUESTION AND ANSWER BOOKLET

Structure of Booklet

Number of questions	*Number of marks*
6	33

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers, a CAS calculator, a scientific calculator and an A4 double sided sheet of notes.
- Students are NOT permitted to bring into the test room: blank sheets of paper.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** and **teacher** in the space provided above on this page.
- All written responses must be in English.
- Answer in **exact values** unless otherwise specified.
- For questions worth more than 1 mark, appropriate working out must be shown. Drawing a diagram/table may constitute working out.

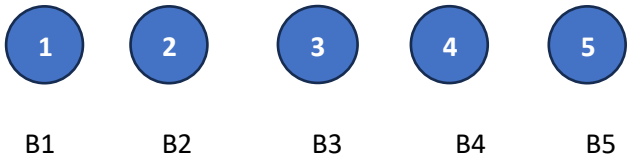
Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Total marks: D1 _____ / 8 D2 _____ / 14 D3 _____/11

Question 1 (9 marks)

Two bags contain different coloured counters.

First bag has five blue counters. Each blue counter shows a different whole number from 1 to 5.



The second bag has three red counters. Each red counter shows a different whole number from 1 to 3.

One counter is drawn from each bag and their numbers are added together.

a. In how many ways can we draw two counters: one red and one blue? (1 mark- D3)

$5 \times 3 = 15$ (1)A

b. List all possible sums (outcomes of this experiment). (1 mark- D1)

$\{2, 3, 4, 5, 6, 7, 8\}$ (1)A

c. List all possible ways in which a sum of 5 can be obtained. (1 mark- D1)

B2+R3, B3+R2, B4+R1 (1)A

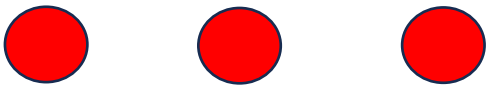
d. Find the probability of getting a sum of 5. (1 mark – D2)

$Pr(\text{sum of } 5) = \frac{3}{15} = \frac{1}{5}$ (1)A

e. Find the probability of getting a sum of at least 5. (2 marks – D2)

$Pr(\text{sum} \geq 5) = 1 - Pr(\text{sum} \leq 4) = 1 - Pr(\text{sum } 2) - Pr(\text{sum} = 3) - Pr(\text{sum} = 4) = 1 - \frac{1}{15} - \frac{2}{15} - \frac{3}{15} = \frac{9}{15} = \frac{3}{5}$ (1)M (1)A

f. By changing the numbers showed on the red counters, make the outcomes of the experiment equally likely. You do not need to show that the outcomes are equally likely. (1 mark – D2)
Write the new numbers on the counters below.



all 3 same number ≥ 6 } a, a, a
a any whole number

or

all 3 different nb.
n, m, p
$n < m < p$
$n \geq 6$ $m \geq n+6$
$p \geq m+6$

or
0, m, p
$m \geq 6$
$p \geq m+6$

Question 2 (8 marks)

$Pr(A \cap B') = 0.24$

Events A and B are such that $\Pr(A) = 0.52$, $\Pr(B) = 0.64$ and $\Pr(A' \cup B) = 0.76$.

a. Construct a probability table. (2 marks – D1) ① → 5 correct ② all correct

	A	A'	Total
B	0.28	0.36	0.64
B'	0.24	0.12	0.36
Total	0.52	0.48	1

b. Find $\Pr(A \cap B')$. (1 mark -D2)

$Pr(A \cap B') = 0.24$ ① A → CM as per table

c. Find $\Pr(A' \cap B')$. (1 mark – D2)

$Pr(A' \cap B') = 0.12$ ① A → CM as per table

d. Find $\Pr(A \cup B)$. (1 mark – D2)

$Pr(A \cup B) = 0.88$ ① A → CM as per table

e. Find $\Pr((A \cap B)')$. (1 mark – D2)

$Pr((A \cap B)') = 0.72$ → CM as per table

f. Find $\Pr(A/B)$. (1 mark – D2)

$Pr(A/B) = \frac{Pr(A \cap B)}{Pr(B)} = \frac{0.28}{0.64} = \frac{7}{16}$ ① A → CM as per table (working shown)

g. Are the events A and B independent? Justify your answer. (1 mark – D2)

$Pr(A/B) = \frac{7}{16} \neq Pr(A) = 0.52 = \frac{52}{100} = \frac{13}{25}$

A and B are not independent events ① A → do not award if not justified

↓ values consistent with the table must be seen

← CM as per table

Do not award if student construct the table using A, B independent as this was not given.

Question 3 (5 marks)

The digits 1,2,3,4,5,6,7,8, and 9 are used to form 3-digit numbers.

a. How many different numbers can we form? (1 mark – D3)

$9 \times 9 \times 9 = 9^3 = 729$ (1)A

b. How many different numbers can we form if digits are not to be repeated? (1 mark – D3)

$9 \times 8 \times 7 = 504$ (1)A

c. When choosing a 3-digit number formed with the digits 1,2,3,4,5,6,7,8, and 9, calculate the probability that the number is even. (2 marks – D3)

→ even no. $4 \times 9 \times 9 = 324$ (1)A

$Pr(\text{even}) = \frac{4 \times \cancel{9} \times \cancel{9}}{9 \times \cancel{9} \times \cancel{9}} = \frac{4}{9}$ (1)A

d. How many different numbers can we form if their digits are in increasing order? (1 mark – D3)

$^9C_3 = 84$ (1)A

Question 4 (4 marks)

A mountain climbing club has 21 members. Out of 21 members, 15 are experienced climbers and 6 are novices. A committee of 5 members must be chosen to organise a social event.

a. Find the total number of different committees that are possible to be formed. (1 mark – D3)

$^{21}C_5 = 20349$ (1)A

b. Find the number of different committees consisting of three experienced climbers and two novices. (1 mark – D3)

$^{15}C_3 \times {}^6C_2 = 6825$ (1)A

c. Find the probability that the committee consists of experiences climbers only. (1 mark – D3)

$Pr(5E) = \frac{^{15}C_5}{^{21}C_5} = \frac{143}{969}$ (1)A

d. Given that at least one experienced climber is in the committee, find the probability that there are more novices in the committee than experienced climbers. (2 marks – D3)

$Pr(\text{more novices} / \text{at least one experienced}) = \frac{Pr(4N+1E) + Pr(3N+2E)}{Pr(5E) + Pr(4E+1N) + Pr(3E+2N) + Pr(2E+3N) + Pr(E+4N)}$ → (1)M

$= \frac{Pr(4N+1E) + Pr(3N+2E)}{1 - Pr(5N)} = \frac{\frac{^6C_4 \times {}^{15}C_1}{^{21}C_5} + \frac{^6C_3 \times {}^{15}C_2}{^{21}C_5}}{1 - \frac{^6C_5}{^{21}C_5}} = \frac{775}{6781}$ (1)A

no mark if $\frac{775}{6783}$ as did not consider conditional prob.

only one mark if 6783 is given instead of 6781

Question 5 (6 marks)

A factory produces 1500 electronic components per day. The components are produced by two different machines.

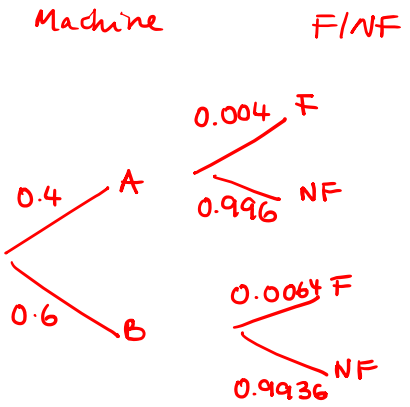
- Machine A produces 600 components each day with an average of 0.4% faulty.
- Machine B produces 900 components each day with an average of 0.64% faulty.

a. What daily percentage of electronic components are produced by Machine A? (1 mark – D1)

$\frac{600}{1500} \times 100 = 40\%$

A component has been chosen at random from the daily output. It is then noted whether the component was faulted or not.

b. Use a tree diagram to represent the information above (2 marks – D1)



(1) → structure A, B & F, NF

(1) probabilities on the branches

(accept % → not ideal)

c. Show that the average percentage of faulty components produced each day is 0.544%. (1 mark – D1)

$Pr(\text{faulty}) = Pr(A \cap F) + Pr(B \cap F) = 0.4 \times 0.004 + 0.6 \times 0.0064$ (1)A must be seen

$= 0.00544$

d. A faulty component has been chosen at random from the combined daily output. Find the probability that the component was produced by Machine A. Give your answer correct to four decimal places. (1 mark – D2)

$Pr(\text{faulty from A} \mid \text{faulty}) = \frac{Pr(A \cap F)}{Pr(\text{faulty})} = \frac{0.4 \times 0.004}{0.00544} \approx 0.2941$ (4 dp) (1)A

e. If Machine B continues to produce 900 components each day, how many should Machine A produce to ensure that the probability of selecting a faulty component from the combined daily output is less than 0.5%? (1 mark – D2)

Let A produce n components $(n > 600)$

$\frac{n}{900+n} \times 0.004 + \frac{900}{900+n} \times 0.0064 < 0.005$

(1)A

$n > 1260$ at least 1261 components

accept 1261 only if given. Do not accept 1260 or > 1260

Question 6 (2 marks – D2)

Two independent events A and B are such that $\Pr(A) = \frac{4}{3} \times \Pr(B)$ and $\Pr(A \cup B) = 0.58$. Find $\Pr(A)$.

$\Pr(A) = p$

$\Pr(B) = \frac{3}{4}p$

	A	A'	
B	$\frac{3}{4}p^2$	$\frac{3}{4}p(1-p)$	$\frac{3}{4}p$
B'	$p(1-\frac{3}{4}p)$	$(1-p)(1-\frac{3}{4}p)$	$1-\frac{3}{4}p$
	p	$1-p$	1

→ 1 M

$1-(1-p)(1-\frac{3}{4}p) = 0.58$

$p = 0.4$

or

$p = 1.933.. > 1$

$p = 0.4$ → 1 A

solve$\left(1-(1-x)\cdot\left(1-\frac{3\cdot x}{4}\right)=0.58, x\right)$

$x=0.4$ or $x=1.93333$

End of assessment